SYNOPSIS

Title of the Proposed Research

Sustainable Chicken Feather-Derived Keratin/Graphene Oxide (GO)/g-C_3N_4 Composite for Visible-Light-Driven Photocatalytic Degradation of Organic Dyes.

Introduction

The rapid growth of the poultry, textile, leather, and related industries has generated substantial amounts of keratin-rich animal fiber waste, particularly chicken feathers, which are often improperly discarded and pose significant environmental challenges [1]. Keratin is a naturally abundant, sulfur-containing fibrous protein possessing various functional groups, including amino, carboxyl, hydroxyl, and thiol/disulfide-related groups, which provide active sites for interacting with organic pollutants. Therefore, converting keratin-rich animal fiber waste into functional materials represents a promising approach for both waste valorization and environmental remediation [2]. At the same time, the discharge of persistent synthetic dyes such as methylene blue, methyl orange, rhodamine B, and Congo red from the textile, printing, leather, and pharmaceutical industries has become a serious water pollution concern because these dyes are chemically stable, resistant to biodegradation, and potentially toxic to aquatic ecosystems. Although conventional treatment techniques such as adsorption, coagulation, filtration, and biological treatment can remove dyes from wastewater, they often transfer pollutants from one phase to another rather than completely degrading them [3]. Thus, integrating the adsorption capability of keratin with photocatalytic technology provides an attractive strategy for efficient dye removal and degradation. Graphitic carbon nitride (g-C_3N_4), a visible-light-responsive semiconductor, can provide photocatalytic activity, while graphene oxide (GO), with its high surface area and oxygen-containing functional groups, can enhance pollutant adsorption and facilitate interfacial charge transfer. Similarly, the development of keratin/GO/g-C_3N_4 ternary composite places renewable animal-fiber-derived keratin at the core of the material design while utilizing GO and g-C_3N_4 as functional components to improve surface properties, charge separation, and visible-light-driven photocatalytic degradation [4], [5], [6]. This approach can transform abundant keratin-rich animal waste into a value-added, sustainable material for the remediation of dye-contaminated water.

In our research, a sustainable and eco-friendly approach can be employed to utilize waste chicken feathers as a renewable source of keratin for developing a keratin/GO/g-C_3N_4 ternary composite for visible-light-driven photocatalytic degradation of organic dyes. Keratin serves as the primary animal-fiber component, providing abundant functional groups and adsorption sites, while graphene oxide (GO) enhances surface area and interfacial charge-transfer properties. Graphitic carbon nitride (g-C_3N_4) contributes visible-light photocatalytic activity, making the composite a promising material for treating dye-contaminated wastewater. The keratin/GO/g-C_3N_4 composite combines adsorption and photocatalytic degradation, where the functional groups of keratin and GO facilitate dye adsorption through surface interactions, while g-C_3N_4 promotes charge generation and reactive species formation under visible-light irradiation. This synergistic combination is expected to improve dye removal efficiency while simultaneously converting abundant chicken-feather waste into a value-added material for sustainable wastewater treatment.

Objectives of the Proposed Research

1. To extract keratin fiber from raw chicken feathers and combine it with GO and g-C_3N_4 to fabricate a ternary composite
2. To investigate the adsorption behavior of dyes on the keratin/GO/g-C_3N_4 ternary composite surface.
3. To characterize the structural, morphological, and optical properties of the keratin/GO and the keratin/GO/g-C_3N_4 ternary composite using techniques such as XRD, SEM, FTIR, and UV-Vis spectroscopy.
4. To evaluate conventional dye degradation using the keratin/GO nanocomposite and compare the effectiveness of the g-C_3N_4 photocatalyst in enhancing dye degradation performance.
5. To perform a statistical analysis evaluating different composites (with photocatalyst & without photocatalyst) at different compositions.
6. To evaluate the adsorption contribution separately using pseudo-first-order and pseudo-second-order kinetic models along with Langmuir and Freundlich isotherms.

Methodology

Graphene oxide (GO) will first be synthesized from graphite powder through a modified chemical oxidation method [7]. Graphite will be treated with an acidic oxidizing mixture, followed by gradual addition of potassium

permanganate under controlled cooling and stirring. The mixture will then be diluted with deionized water and heated to promote oxidation and exfoliation. Hydrogen peroxide will be added to terminate the reaction, followed by washing with dilute HCl and deionized water to remove impurities. The resulting GO will be separated and dried for further use [7]. On the other hand, Keratin will be extracted and purified from waste chicken feathers using an appropriate reduction/solubilization process, followed by washing, drying, and pretreatment of the feathers [5]. The extracted keratin will then be combined with graphene oxide (GO), produced from raw graphite powder, and commercially procured graphitic carbon nitride (g-C_3N_4) at different ratios to fabricate the keratin/GO/g-C_3N_4 ternary composite [8]. The prepared materials will be characterized using FTIR, XRD, SEM-EDX, TEM, BET, UV-Vis spectroscopy, and photoluminescence (PL) analysis to investigate their functional groups, structural properties, morphology, surface characteristics, optical behavior, and charge-carrier recombination. Different single-component, binary, and ternary systems will be tested under identical conditions to determine the contribution of each component and identify the optimum composite composition.

Photocatalytic removal study

The visible-light-driven photocatalytic degradation of selected dyes, particularly methylene blue and methyl orange, will be investigated using the optimized keratin/GO/g-C_3N_4 composite. Key operating parameters, including solution pH, photocatalyst dosage, initial dye concentration, irradiation time, and light intensity, will be optimized to obtain maximum degradation efficiency. Before irradiation, the dye-catalyst suspension will be stirred in the dark to establish adsorption-desorption equilibrium. During visible-light irradiation, samples will be collected at fixed time intervals, separated from the photocatalyst, and analyzed using UV-Vis spectrophotometry. The adsorption contribution will be evaluated separately using pseudo-first-order and pseudo-second-order kinetic models along with Langmuir and Freundlich isotherms, while photocatalytic degradation will primarily be analyzed using the pseudo-first-order kinetic model. Radical-scavenger experiments will further be conducted to identify the major reactive species involved in dye degradation [6].

Expected Results

1. Successful extraction of keratin from chicken feathers and effective incorporation of GO and g-C_3N_4, confirmed through structural, morphological, and spectroscopic characterization.
2. The optimized keratin/GO/g-C_3N_4 ternary composite is expected to exhibit enhanced dye adsorption and superior visible-light-driven photocatalytic degradation compared with pristine g-C_3N_4 and the corresponding binary composites.
3. Improved charge separation and reduced electron-hole recombination are expected, supported by photoluminescence and reactive-species studies, resulting in enhanced photocatalytic performance.
4. The optimized composite is expected to retain significant photocatalytic activity after repeated regeneration cycles, demonstrating its potential as a sustainable and reusable material for wastewater treatment.

Time Frame

The proposed research will be designed for 1 year.

Activity	Month 1-2	Month 3-4	Month 5-6	Month 7-8	Month 9-10	Month 11-12
Collection and Preparation of Materials						
Extraction and Optimization						
Characterization						
Application Study						
Structural and Environmental Analysis						
Data Analysis and Final Report Preparation						

Significance of the Project

The proposed research aims to convert abundant chicken-feather waste into a value-added functional material for wastewater treatment in Bangladesh. The developed keratin/GO/g-C_3N_4 ternary composite is expected to combine the adsorption capability of keratin and GO with the visible-light photocatalytic activity of g-C_3N_4, enabling both adsorption and degradation of persistent organic dyes. This approach may provide an environmentally friendly, sustainable, and potentially reusable alternative for treating dye-contaminated wastewater while simultaneously addressing poultry waste disposal. Furthermore, the study may contribute to the development of low-cost bio-derived photocatalytic materials with improved charge separation and visible-light-driven degradation performance.

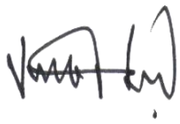
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